Milestone 2 Report

Smart MCQ Solver Challenge

Deep Learning and Generative AI Project

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1. Introduction

Milestone 2 extends the classical NLP baseline developed during Milestone 1 by introducing Transformer-based Natural Language Processing techniques. Unlike TF-IDF and Word2Vec, transformer models generate contextual representations that capture semantic relationships between words and sentences. The milestone focuses on understanding the Hugging Face ecosystem, BERT architecture, contextual embeddings, attention mechanisms, zero-shot classification, and small language models.

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2. Objectives

The objectives of this milestone are:

- Learn the Hugging Face datasets library.
- Understand the Hugging Face transformers ecosystem.
- Study BERT tokenization.
- Learn special tokens.
- Understand self-attention.
- Explore contextual embeddings.
- Compare TF-IDF with transformer embeddings.
- Perform zero-shot classification.
- Experiment with small language models.

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3. Hugging Face Datasets

The dataset was loaded using the Hugging Face datasets library instead of pandas.

The `.map()` function was used to generate combined textual inputs consisting of:

- Question prompt
- Option A

- Option B
- Option C
- Option D
- Option E

This preprocessing method allows efficient dataset transformations while preserving metadata.

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4. Tokenization

The `bert-base-uncased` tokenizer was initialized using the Hugging Face Transformers library.

The tokenizer performs:

- Lowercasing
- WordPiece tokenization
- Vocabulary lookup
- Special token insertion

Batch tokenization was applied to the entire dataset.

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5. Special Tokens

Several special tokens were explored:

- `[CLS]`
- `[SEP]`
- `[PAD]`
- `[MASK]`
- `[UNK]`

Each token serves a specific purpose during transformer inference and training.

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6. BERT Architecture

The milestone examined the internal architecture of the BERT encoder.

Key architectural components include:

- Hidden size = 768

- Number of attention heads = 12
- Multi-layer Transformer encoder
- Feed-forward neural networks
- Residual connections
- Layer normalization

The attention head dimension was also calculated.

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7. Contextual Embeddings

Unlike TF-IDF vectors, transformer embeddings depend on surrounding words.

Context-aware sentence embeddings were generated using the Sentence Transformers library with the `all-MiniLM-L6-v2` model.

These embeddings capture semantic meaning rather than simple word frequency.

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8. TF-IDF vs Transformer Embeddings

A comparison between classical and transformer-based representations was performed.

TF-IDF

- Sparse vectors
- Frequency based
- No contextual information

Sentence Transformers

- Dense vectors
- Context-aware
- Better semantic understanding
- Higher retrieval quality

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9. Zero-Shot Classification

Zero-shot classification was implemented using the `facebook/bart-large-mnli` model.

Both standard inference and `multi_label=True` inference were explored.

This demonstrates how transformer models can classify unseen labels without additional task-specific training.

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10. Small Language Models

The milestone concludes with experiments using the `google/flan-t5-small` model.

The model was used for instruction-following text generation and demonstrates the transition from discriminative NLP models toward generative AI systems.

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11. Results

The milestone successfully demonstrated:

- Dataset loading using Hugging Face.
- Efficient preprocessing with `.map()`.
- Transformer tokenization.
- BERT architecture understanding.
- Context-aware sentence embeddings.
- TF-IDF vs Transformer comparison.
- Zero-shot classification.
- Small language model inference.

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12. Challenges

Several implementation challenges were encountered:

- Library version compatibility.
- Managing transformer dependencies.
- GPU memory usage.
- Token length limitations.
- Differences between encoder and generative models.

These challenges provided practical experience in working with transformer-based NLP pipelines.

14. Conclusion

Milestone 2 successfully introduced transformer-based natural language processing techniques. Through experiments with Hugging Face datasets, BERT tokenization, contextual embeddings, Sentence Transformers, zero-shot classification, and FLAN-T5, a strong foundation has been established for building more advanced retrieval and reasoning systems